MIL-C-62499(AT)
1 April 1985
SUPERSEDING
(see 6.4)

# MILITARY SPECIFICATION

# CUSHION ASSEMBLIES

This specification is approved for use by US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers cushion assemblies used as seats, backrests, and headrests in tracked combat vehicles.

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GSS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 2540

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SPECIFICATIONS
FEDERAL

| | |
|---|---|
| L-P-386 | - Plastic Material, Cellular, Urethane (Flexible). |
| P-C-437 | - Cleaning Compound, High Pressure (Steam) Cleaner. |
| VV-F-800 | - Fuel Oil, Diesel. |
| CCC-C-700 | - Cloth, Coated, Vinyl Coated (Artificial Leather). |

MILITARY

| | |
|---|---|
| MIL-G-3056 | - Gasoline, Automotive, Combat. |
| MIL-T-5624 | - Turbine Fuel, Aviation, Grades JP-4 and JP-5. |
| MIL-H-6083 | - Hydraulic Fluid, Petroleum Base, for Preservation and Testing. |
| MIL-F-13927 | - Fungus Resistance Test, Automotive Components. |
| MIL-P-14232 | - Parts, Equipment and Tools for Army Materiel, Packaging and Packing of. |
| MIL-F-16884 | - Fuel Oil, Diesel, Marine. |
| MIL-H-46170 | - Hydraulic Fluid, Rust Inhibited, Fire-Resistant Synthetic Hydrocarbon Base. |

STANDARDS
FEDERAL

| | |
|---|---|
| FED-STD-191 | - Textile Test Methods. |

MILITARY

| | |
|---|---|
| MIL-STD-105 | - Sampling Procedures and Tables for Inspection by Attributes. |
| MIL-STD-129 | - Marking for Shipment and Storage. |
| MIL-STD-810 | - Environmental Test Methods and Engineering Guidelines. |
| MIL-STD-45662 | - Calibration System Requirements. |

NATIONAL HIGHWAY TRAFFIC SAFETY ADMINISTRATION

| | |
|---|---|
| FMVSS 302 | - Flammability of Interior Materials - Passenger Cars, Multipurpose Passenger Vehicles, Trucks, and Buses. |

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

| | |
|---|---|
| ASTM D 429 | - Rubber Property - Adhesion to Rigid Substrates, Test Methods for. |
| ASTM D 751 | - Coated Fabrics, Methods of Testing. |
| ASTM D 1790 | - Brittleness Temperature of Plastic Sheeting by Impact, Test Method for. |
| ASTM D 3574 | - Flexible Cellular Materials. |

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA, 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), the contractor shall furnish cushion assemblies which shall be subjected to first article inspection (see 4.4 and 6). First article inspection samples, properly marked with identifying information shall be representative of the unit to be furnished to the Government. All subsequent cushion assemblies delivered to the Government shall conform to these samples in all of their pertinent physical and performance attributes.

3.2 Materials. The materials shall be as specified herein. The cushion assemblies shall not contain free toluene diisocyanate or any other materials which would cause discomfort or adverse effects on the health of the personnel (see 4.6.1).

3.2.1 Padding. The padding shall consist of a foamed-in-place cellular elastomeric material conforming to type V, grade HR-II of L-P-386 (see 4.6.1).

3.2.2 Cover material. (This space intentionally left blank.)

3.3 Design and construction.

3.3.1 Configuration. The cushion assembly shall conform to the applicable assembly drawing (see 4.5.2 and 4.6.2).

3.3.2 Color. The color of the cushion assembly shall be black (see 4.5.2 and 4.6.2).

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3.3.3 Cover material weight. The cover material shall have a total weight of 44 ± 2 ounces per square yard with the fabric portion being 5.8 ounces per square yard (see 4.6.1).

3.3.4 Cover material gauge. The total gauge of the cover material shall be 74 ± 7 mil with the skin thickness being 30 ± 1 mil (see 4.6.1).

3.4 Performance.

3.4.1 Cushion assembly.

3.4.1.1 Load capacity. The cushion assembly shall retain 90 percent of its original dimensions after compression for 24 hours with 350 pounds of force distributed over an area 11 inches in diameter (see 4.6.3.1.1).

3.4.1.2 Impact resistance. The cushion assembly shall withstand 25 000 impacts of 57 ± 2 pound feet applied to an area 8 inches in diameter without flattening the padding, tearing the cover material, or deforming the bottom (see 4.6.3.1.2).

3.4.2 Padding.

3.4.2.1 Tear resistance. The padding shall have a tear resistance of not less than 1 pound force per inch (see 4.6.3.2.1).

3.4.3 Adhesive.

3.4.3.1 Adhesion strength. The adhesive bonding material shall have an adhesion strength of at least 20 pounds force per square inch (psi) (see 4.6.3.3.1).

3.4.4 Cover material.

3.4.4.1 Stability. The exposed surfaces of the cover material shall not become tacky or sticky, emit odors, or stain clothing (see 4.6.1).

3.4.4.1.1 Surface condition. The cover material shall be free of cracks and checks (see 4.5.2 and 4.6.2).

3.4.4.2 Tensile strength (warp and filler). The tensile strength of the cover material shall be at least 160 pounds force per inch (see 4.6.3.4.1).

3.4.4.3 Tear strength (warp and filler). The tear strength of the cover material shall be at least 60 pounds (see 4.6.3.4.2).

3.4.4.4 Coating adhesion strength. The coating of the cover material shall have an adhesion strength of at least 10 pounds force per inch (see 4.6.3.4.3).

3.4.4.5 Cold impact. When tested as specified in 4.6.3.4.4, the cover material shall not exhibit a failure at -40 degrees Fahrenheit (°F).

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3.4.4.6 Color fastness. When tested as specified in 4.6.3.4.5, the cover material shall have a rating of at least "good", with no surface degradation.

3.4.4.7 Wear resistance. When tested as specified in 4.6.3.4.6, the cover material shall show no evidence of wearing through.

3.4.4.8 Crocking. When tested as specified in 4.6.3.4.7, the cover material shall show no evidence of crocking.

3.4.4.9 Volatility. When tested as specified in 4.6.3.4.8, the cover material shall have no more than a 6 percent weight loss.

3.4.4.10 Grain retention. When tested as specified in 4.6.3.4.9, the cover material shall show no evidence of loss of grain.

3.4.4.11 Stretch. (This space intentionally left blank.)

3.4.4.12 Flame resistance. The cushion assembly shall meet the requirements of Federal Motor Vehicle Safety Standard (FMVSS) No. 302 (see 4.6.3.4.11).

3.4.5 Environmental.

3.4.5.1 Cushion assembly.

3.4.5.1.1 High temperature compression set. The cushion assembly shall not exceed a compression set of 20 percent when tested as specified in 4.6.3.5.1.1.

3.4.5.1.2 Low temperature compression set. The cushion assembly shall not exceed a compression set of 25 percent when tested as specified in 4.6.3.5.1.2.

3.4.5.1.3 High temperature. The cushion assembly shall retain at least 90 percent of its original dimensions when tested as specified in 4.6.3.5.1.3.

3.4.5.1.4 Low temperature. The cushion assembly shall retain at least 90 percent of its original dimensions when tested as specified in 4.6.3.5.1.4.

3.4.5.1.5 Humidity. The cushion assembly shall meet the requirements of 3.4.1.1 after exposure to 100 percent relative humidity (see 4.6.3.5.1.5).

3.4.5.1.6 Fungus. The cushion assembly shall meet the requirements of 3.4.1.1 after exposure to conditions favorable to fungus growth. These conditions shall be as defined in MIL-F-13927 (see 4.6.3.5.1.6).

3.4.5.1.7 Submergence. The cushion assembly shall meet the requirements of 3.4.1.1 after submergence in water to a depth of 1 inch for at least 30 minutes (see 4.6.3.5.1.7).

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3.4.5.1.8 Chemicals. The cushion assembly shall meet the requirements of 3.4.1.1 after exposure to the vapors of and contact with the following chemicals (see 4.6.3.5.1.8):

a. Fuel per VV-F-800, MIL-T-5624, MIL-G-3056, and MIL-F-16884.
b. Hydraulic fluid per MIL-H-6083 and MIL-H-46170.
c. Cleaning agents per P-C-437.

3.4.5.1.9 Waterjet spraying. The cushion assembly shall meet the requirements of 3.4.1.1 after, and shall not be damaged by, waterjet spraying as specified in 4.6.3.5.1.9.

3.5 Identification and marking. Identification and marking shall be as specified on applicable drawings.

3.6 Workmanship. Cushion assemblies shall be free from cracks or porosity in the cover material. The sides shall be free of seams and laps. Surfaces shall be smooth and free of bumps, blisters, or abrasions. The supporting pan shall be free of sharp edges or corners.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or witness any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection equipment. Unless otherwise specified in the contract or purchase order (see 6.2), the supplier is responsible for the provision and maintenance of all inspection equipment necessary to assure that supplies and services conform to contract requirements. Commercial, modified commercial, or supplier designed inspection equipment or measuring set-ups must be capable of repetitive measurements to an accuracy of 10 percent of the component tolerance. Calibration of inspection equipment shall be in accordance with MIL-STD-45662. Where compliance with MIL-STD-45662 is not possible, special calibration requirements shall be prepared for approval by the Government.

4.2 Classification of inspection:

a. First article inspection (see 4.4).
b. Quality conformance inspections (see 4.5).
   1. Examination (see 4.5.2).
   2. Acceptance tests (see 4.5.3.1).
   3. Control tests (see 4.5.3.2).

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4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following conditions:

a. Air temperature 80 ± 10°F
c. Relative humidity 35 ± 5 percent

4.4 First article inspection. First article inspections shall be performed on preproduction or initial production samples as specified herein. Approval of the first article sample by the Government shall not relieve the contractor of the obligation to supply cushion assemblies that are fully representative of those inspected as a first article sample. Any changes or deviation of the production units from the first article sample shall be subject to the approval of the contracting officer.

4.4.1 Preproduction inspection. When specified (see 6.2), the preproduction sample shall consist of 2 cushion assemblies. Preproduction units shall be numbered and inspected in the sequence of table I and as specified in table II.

4.4.2 Initial production inspection. Unless otherwise specified (see 6.2), the Government shall select 2 units from the first 10 cushion assemblies produced under the production contract for initial production inspection. Initial production units shall be numbered and inspected in the sequence of table I and as specified in table II.

4.4.3 First article inspection failure. Test item deficiencies during, or as a result of, the first article test, shall be cause for rejection of the items until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiency. Any deficiency found during, or as a result of the first article test, shall be prima facie evidence that all items already produced prior to completion of the first article test are similarly deficient unless evidence satisfactory to the contracting officer is furnished by the contractor that they are not similarly deficient. Such deficiencies on all items shall be corrected by the contractor at no cost to the Government. The Government shall not final accept products until first article testing is completed to the satisfaction of the Government.

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TABLE I. First article sequence.

| Test sequence | Paragraph | First article sample | |
|---|---|---|---|
| | | 1 | 2 |
| Load capacity | 4.6.3.1.1 | X | X |
| Humidity | 4.6.3.5.1.5 | | X |
| Fungus | 4.6.3.5.1.6 | X | |
| Submergence | 4.6.3.5.1.7 | X | |
| Chemicals | 4.6.3.5.1.8 | | X |
| Low temperature | 4.6.3.5.1.4 | X | X |
| Low temperature compression set | 4.6.3.5.1.2 | X | |
| High temperature | 4.6.3.5.1.3 | X | X |
| High temperature compression set | 4.6.3.5.1.1 | | X |
| Cleaning spray | 4.6.3.5.1.9 | X | |
| Impact resistance | 4.6.3.1.2 | X | X |

TABLE II. Classification of inspections.

| Title | Requirement | Inspection | First article | Quality conformance Acceptance |
|---|---|---|---|---|
| Materials and construction | 3.2, 3.2.1, 3.2.2, 3.3.3, 3.3.4 and 3.4.4.1 | 4.6.1 | X | |
| Defects (see table III) | 3.3.1, 3.3.2 and 3.4.4.1.1 | 4.6.2 | X | |
| Load capacity | 3.4.1.1 | 4.6.3.1.1 | X | X |
| Impact resistance | 3.4.1.2 | 4.6.3.1.2 | X | |
| Tear resistance | 3.4.2.1 | 4.6.3.2.1 | X | |
| Adhesion strength | 3.4.3.1 | 4.6.3.3.1 | X | |
| Tensile strength | 3.4.4.2 | 4.6.3.4.1 | X | |
| Tear strength | 3.4.4.3 | 4.6.3.4.2 | X | |
| Coating adhesion strength | 3.4.4.4 | 4.6.3.4.3 | X | |
| Cold impact | 3.4.4.5 | 4.6.3.4.4 | X | |
| Colorfastness | 3.4.4.6 | 4.6.3.4.5 | X | |
| Wear resistance | 3.4.4.7 | 4.6.3.4.6 | X | |
| Crocking | 3.4.4.8 | 4.6.3.4.7 | X | |
| Volatility | 3.4.4.9 | 4.6.3.4.8 | X | |
| Grain retention | 3.4.4.10 | 4.6.3.4.9 | X | |
| Stretch | 3.4.4.11 | 4.6.3.4.10 | X | |
| Flame resistance | 3.4.4.12 | 4.6.3.4.11 | X | |

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TABLE II. Classification of inspections.

| Title | Requirement | Inspection | First article | Quality conformance Acceptance |
|---|---|---|---|---|
| High temperature compression set | 3.4.5.1.1 | 4.6.3.5.1.1 | X | |
| Low temperature compression set | 3.4.5.1.2 | 4.6.3.5.1.2 | X | |
| High temperature | 3.4.5.1.3 | 4.6.3.5.1.3 | X | |
| Low temperature | 3.4.5.1.4 | 4.6.3.5.1.4 | X | |
| Humidity | 3.4.5.1.5 | 4.6.3.5.1.5 | X | |
| Fungus | 3.4.5.1.6 | 4.6.3.5.1.6 | X | |
| Submergence | 3.4.5.1.7 | 4.6.3.5.1.7 | X | |
| Chemicals | 3.4.5.1.8 | 4.6.3.5.1.8 | X | |
| Waterjet spray | 3.4.5.1.9 | 4.6.3.5.1.9 | X | |
| Inspection of packaging | 5.1 | 4.6.4 | | X |

4.5 Quality conformance inspection.

4.5.1 Sampling.

4.5.1.1 Lot formation. An inspection lot shall consist of all cushion assemblies of one type and size, from an identifiable production period, submitted at one time for acceptance.

4.5.1.2 Sampling for examination. Samples for quality conformance examination shall be selected in accordance with general inspection level II of MIL-STD-105.

4.5.1.3 Sampling for acceptance. Samples for acceptance tests shall be selected in accordance with inspection equipment level S-3 of MIL-STD-105.

4.5.2 Quality conformance examinations.

4.5.2.1 Acceptable quality level. Each sample selected in accordance with 4.5.1.2 shall be examined to determine conformance to the following acceptable quality levels (AQL) on the basis of percent defective (see 6.4):

| Classification | AQL |
|---|---|
| Major | 2.5 |
| Minor | 4.0 |

4.5.2.2 Classification of defects. For examination purposes, defects shall be classified as listed in table III.



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TABLE III. Classification of defects.

| Category | Defect | Method of examination |
|---|---|---|
| Critical | None | |
| Major | AQL 2.5% Defective | |
| 101 | Cushion assembly, incomplete (see 3.3). | Visual |
| 102 | Dimensions affecting interchangeability, out of tolerance (see 3.3.1). | SIE 1/ |
| 103 | Color not as specified (see 3.3.2). | Visual |
| 104 | Cover material, cracked or checked (see 3.4.4.1.1). | Visual |
| 105 | Identification marking, improper (see 3.5). | Visual |
| 106 | Faulty workmanship affecting performance (see 3.6). | Visual |
| Minor | AQL 4.0% Defective | |
| 201 | Dimensions not affecting interchangeability, out of tolerance (see 3.3.1). | SIE 1/ |
| 202 | Faulty workmanship affecting appearance (see 3.6). | Visual |

1/ SIE = Standard Inspection Equipment.

4.5.3 Quality conformance tests.

4.5.3.1 Acceptance tests. Samples selected in accordance with 4.5.1.3 shall be subjected to the tests specified in table II.

4.5.3.2 Failure. Failure of any cushion assembly to pass any of the specified inspections shall be cause for the Government to refuse acceptance of the production quantity represented, until action taken by the contractor to correct defects and prevent recurrence has been approved by the Government.

4.6 Methods of inspection.

4.6.1 Materials and construction. To determine conformance to 3.2, 3.2.1, 3.2.2, 3.3.3, 3.3.4 and 3.4.4.1, inspection and material certification records shall be maintained by the contractor. Records shall be subject to review by the Government and shall include date, part, or characteristic identification, inspection results, and disposition of lot (accepted or rejected). Corrective action taken on noted defects shall be subject to approval by the Government.

4.6.2 Defects. Conformance to 3.3.1, 3.3.2 and 3.4.4.1.1 shall be determined by examination for the defects listed in table II. Examination shall be visual, tactile, or by measurement with standard inspection equipment.

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4.6.3 Performance.

4.6.3.1 Cushion assembly.

4.6.3.1.1 Load capacity. To determine conformance to 3.4.1.1, a test weight of 350 pounds applied over a distribution plate with a base diameter of 11 ± 1 inches shall be applied to the cushion assembly for 24 hours. At the conclusion of this load application period, remove the test weight and plate and, after 15 minutes, measure and examine the cushion assembly for retention of original shape and dimensions.

4.6.3.1.2 Impact resistance. To determine conformance to 3.4.1.2, subject the cushion assembly to 25 000 impacts of 57 ± 2 pound feet with a 8 ± 1 inch diameter striker with a radius of curvature of 16 inches. Apply impacts at the rate of 20 per minute through a swivel joint so that the load will be applied over the 8 inch diameter regardless of the angle of the top surface of the test specimen. At the conclusion of this test visually examine for evidence of flattening the padding, tearing the cover material, or deforming the bottom.

4.6.3.2 Padding.

4.6.3.2.1 Tear resistance. To determine conformance to 3.4.2.1, the padding shall be tested in accordance with ASTM D 3574.

4.6.3.3 Adhesive.

4.6.3.3.1 Adhesion strength. To determine conformance to 3.4.3.1, the adhesive bonding material shall be tested in accordance with method B of ASTM D 429.

4.6.3.4 Cover material.

4.6.3.4.1 Tensile strength. To determine conformance to 3.4.4.2, the cover material shall be tested in accordance with ASTM D 751.

4.6.3.4.2 Tear strength. To determine conformance to 3.4.4.3, the cover material shall be tested in accordance with method 5134 of FED-STD-191.

4.6.3.4.3 Coating adhesion strength. To determine conformance to 3.4.4.4, the cover material shall be tested in accordance with ASTM D 751.

4.6.3.4.4 Cold impact. To determine conformance to 3.4.4.5, the cover material shall be tested in accordance with ASTM D 1790.

4.6.3.4.5 Color fastness. To determine conformance to 3.4.4.6, the cover material shall be tested in accordance with method 5660 of FED-STD-191.

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4.6.3.4.6 Wear resistance. To determine conformance to 3.4.4.7, the cover material shall be tested in accordance with method 5306 of FED-STD-191, with the following specifics:

a. Load: 1000 grams.
b. Wheel type: H-18.
c. Revolutions: 4000.

4.6.3.4.7 Crocking. To determine conformance to 3.4.4.8, the cover material shall be tested in accordance with method 5651 of FED-STD-191, with a repetition of 10 double strokes.

4.6.3.4.8 Volatility. The cover material shall be weighed in accordance with CCC-C-700, heat soaked at 220°F for 48 hours, then stabilized at room temperature. The cover material shall then be reweighed to determine conformance to 3.4.4.9.

4.6.3.4.9 Grain retention. A sample of the cover material, at least 5 inches by 5 inches, shall be heat soaked at 275°F for 1 hour. The sample, while flat, shall be examined to determine conformance to 3.4.4.10.

4.6.3.4.10 Stretch. (This space intentionally left blank.)

4.6.3.4.11 Flame resistance. To determine conformance to 3.4.4.12, the cushion assembly shall be tested in accordance with FMVSS 302.

4.6.3.5 Environmental.

4.6.3.5.1 Cushion assembly.

4.6.3.5.1.1 High temperature compression set. To determine conformance to 3.4.5.1.1, the cushion assembly shall be heat soaked at 140°F for 47 hours, then the test weight of 4.6.3.1.1 shall be applied for 1 hour at the 140°F temperature environment. Remove the test weight, allow the cushion assembly to stabilize at room temperature, then measure the thickness and calculate the compression set. (NOTE: The compression set is the difference between the original and final thickness, divided by the original thickness.)

4.6.3.5.1.2 Low temperature compression set. To determine conformance to 3.4.5.1.2, repeat the test of 4.6.3.5.1.1 except the temperature shall be -70°F.

4.6.3.5.1.3 High temperature. To determine conformance to 3.4.5.1.3, the cushion assembly shall be tested in accordance with MIL-STD-810, method 501.2, procedures I and II, with the following specifics:

a. Procedure I: 160°F and 15 percent relative humidity.
b. Procedure II: 140°F for 48 hours and 15 percent relative humidity.

After stabilizing at room temperature, the cushion assembly shall be tested in accordance with 4.6.3.1.1.

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4.6.3.5.1.4 Low temperature. To determine conformance to 3.4.5.1.4, the cushion assembly shall be tested in accordance with MIL-STD-810, method 502.2, procedures I and II, at -70°F for 24 hours. After stabilizing at room temperature, the cushion assembly shall be tested in accordance with 4.6.3.1.1.

4.6.3.5.1.5 Humidity. The cushion assembly shall be tested in accordance with MIL-STD-810, method 507.2, procedure II. The conditions shall conform to figure 507.2-2, cycle 4, except the humidity limits shall be 5 and 100 percent. After stabilizing at the conditions specified in 4.3, the cushion assembly shall be tested in accordance with 4.6.3.1.1 to determine conformance to 3.4.5.1.5.

4.6.3.5.1.6 Fungus. To determine conformance to 3.4.5.1.6, the cushion assembly shall be tested in accordance with MIL-F-13927. In lieu of a test, certification, with supporting data, may be provided attesting that the cushion assembly is constructed of materials that will not support fungus growth.

4.6.3.5.1.7 Submergence. The cushion assembly shall be submerged in water to a depth of 1 inch for at least 30 minutes. (NOTE: Submergence depth shall be measured between the water surface and the top surface of the cushion assembly.) After submergence the cushion assembly shall be allowed to dry for 48 hours. After drying, the cushion assembly shall be tested in accordance with 4.6.3.1.1 to determine conformance to 3.4.5.1.7.

4.6.3.5.1.8 Chemicals. The cushion assembly shall be liberally wetted with the chemicals specified in 3.4.5.1.8, wiped clean, and after 1 hour examined for deterioration, tackiness, or loss of color. The cushion assembly shall then be tested in accordance with 4.6.3.1.1 to determine conformance to 3.4.5.1.8.

4.6.3.5.1.9 Waterjet spray. The cushion assembly shall be subjected to waterjet spraying through a nozzle having an orifice diameter not larger than 0.25 inch with the pressure not more than 25 psi at the rate of 1 square foot per minute. The cushion assembly shall receive 200 percent coverage, then allowed to dry for 16 hours at room temperature. The cushion assembly shall then be examined for damage, and tested in accordance with 4.6.3.1.1 to determine conformance to 3.4.5.1.9.

4.6.4 Inspection of packaging. To determine conformance to 5.1, preservation, packaging, packing, and marking shall be inspected as specified in 4.6.1 and 4.6.2.

## 5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for the desired level shall be in accordance with the applicable packaging standard or packaging data sheet specified by the contracting authority (see 6.2).



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6. NOTES

6.1 Intended use. The cushion assemblies covered by this specification are intended to be used as seats, backrests, and headrests in tracked combat vehicles.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Title, number, and date of applicable drawings.
c. If first article samples are not required (see 3.1).
d. If responsibility for inspection shall be other than as specified (see 4.1).
e. If responsibility for inspection equipment shall be other than as specified (see 4.1.1).
f. If inspection conditions shall be other than as specified (see 4.3).
g. If preproduction inspection is required (see 4.4.1).
h. If initial inspection is not required (see 4.4.2).
i. Applicable levels of preservation, packaging, packing, and marking (see 5.1).

6.3 Recycled materials. The use of recycled materials which meet the requirements of the applicable material specifications without jeopardizing the intended use of the item shall be encouraged.

6.4 Supersession data. This document supersedes General Dynamics specification SC-X15202A, dated 21 June 1982.

Custodian:
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